AI/ML-Based Campus Maintenance Reporting System

System Architecture, Technology Stack & UI Wireframes

Capstone Design Project – INNOVEXA

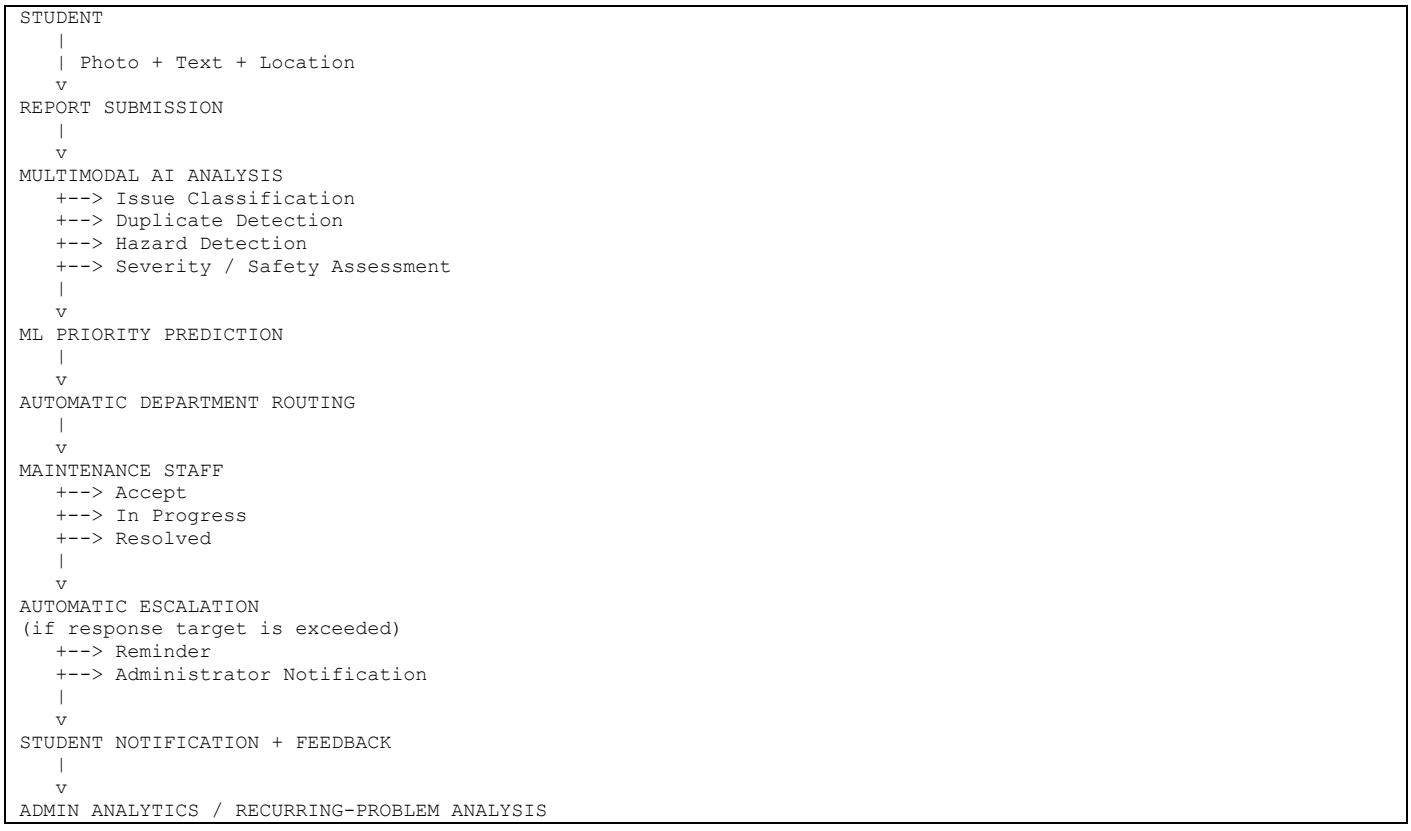
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AI & Computer Engineering Department | 2026 2nd Semester

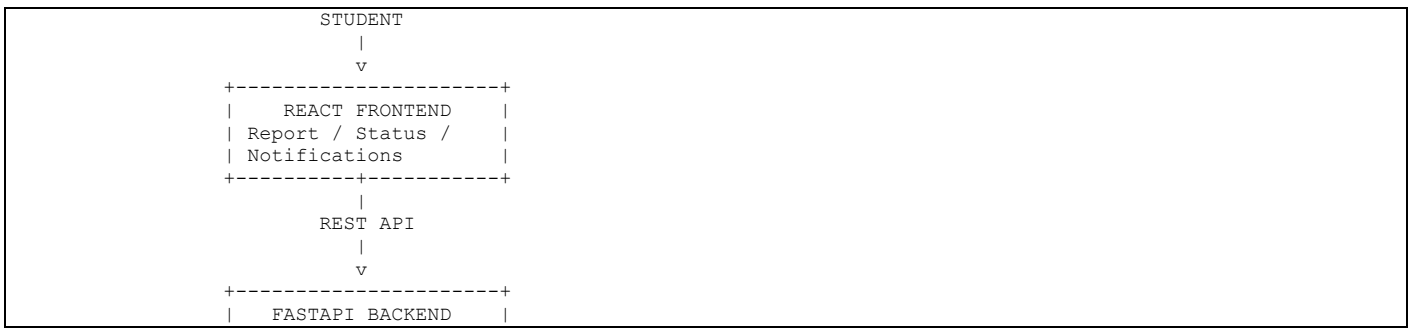
1. Project Overview

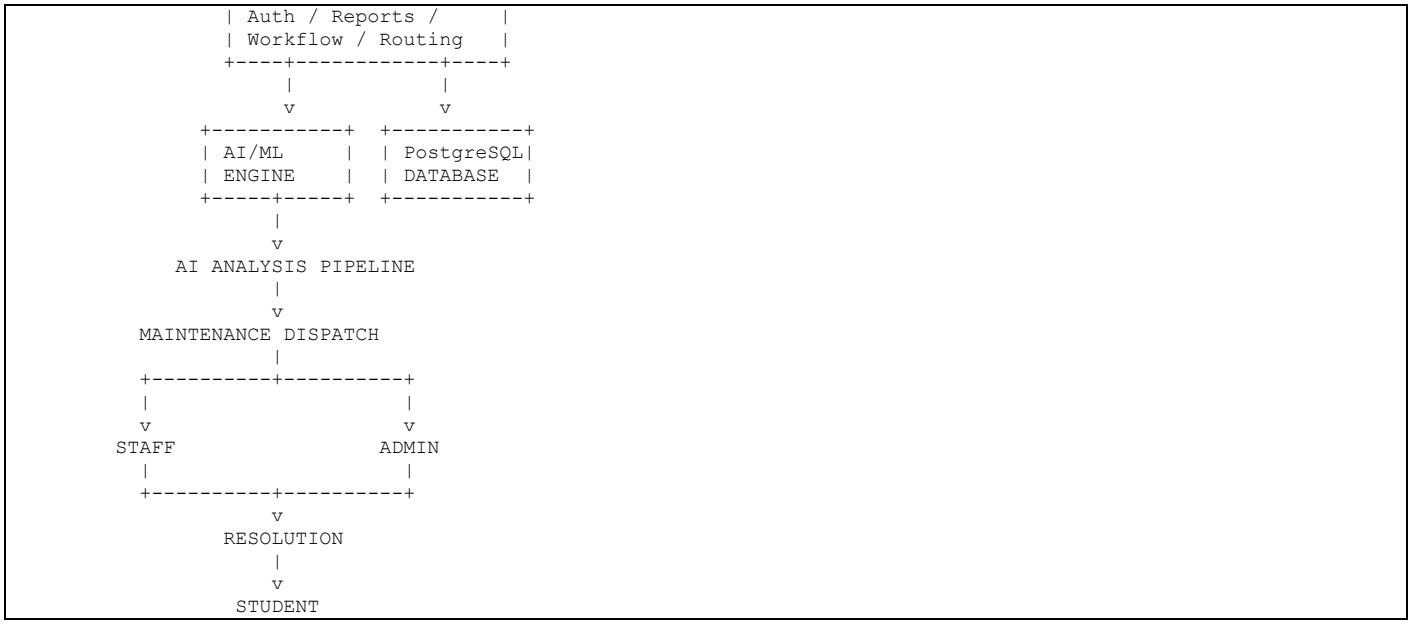
The AI/ML-Based Campus Maintenance Reporting System is a web-based campus facility maintenance platform. Students submit a facility problem using text, an optional photograph, and location information. The system analyzes the report using AI/ML, checks for duplicate incidents, assesses visible safety hazards, predicts maintenance priority, routes the issue to the appropriate maintenance division, and tracks the task through resolution. An automatic escalation module monitors delayed responses, while administrator analytics provide information about recurring problems and maintenance workload.

2. Core Workflow

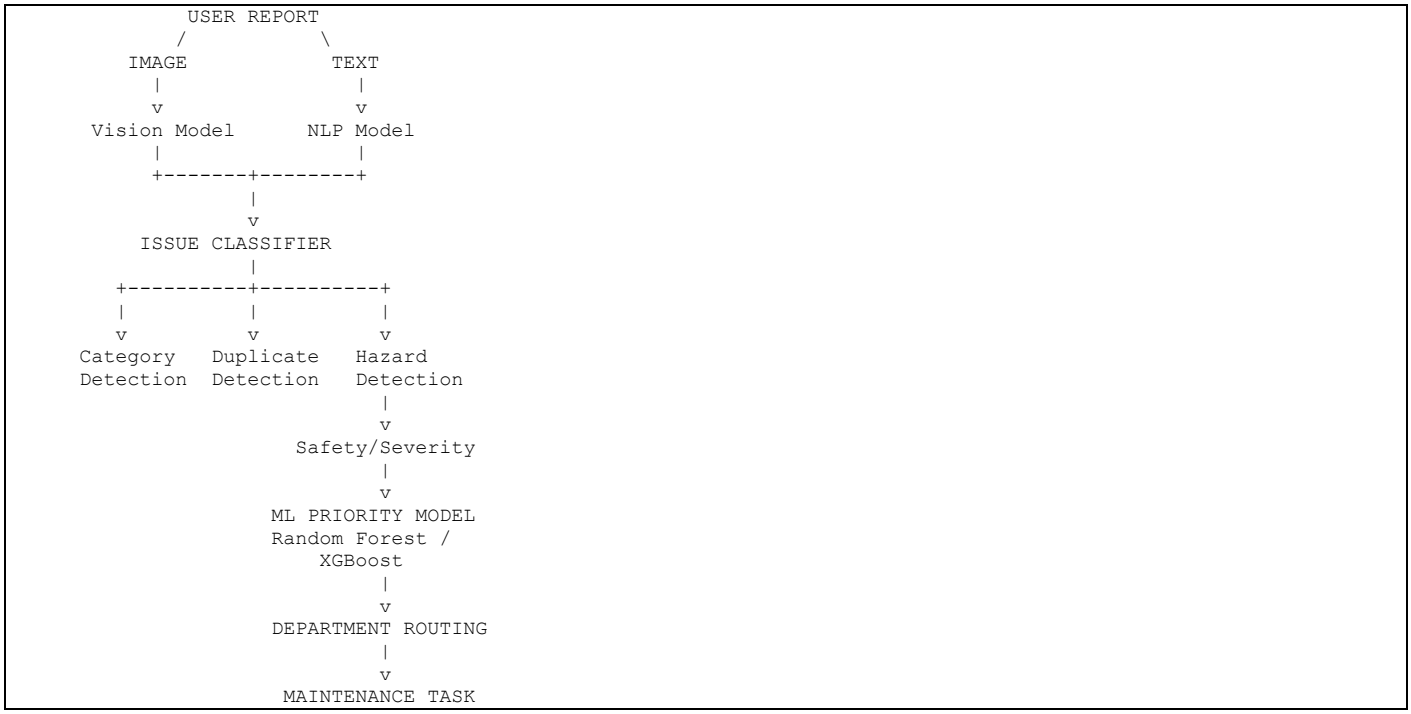


3. High-Level System Architecture





4. AI/ML Processing Architecture



5. Main AI/ML Components

Component	Purpose	Possible Technology
Issue classification	Identify the facility problem category from text and image information.	Multimodal model / CLIP or lightweight VLM; NLP baseline
Duplicate detection	Find reports that may describe the same active physical incident.	Image/text embeddings + cosine similarity
Hazard detection	Identify predefined visible hazards such as exposed wiring, water pooling, cracks, or broken glass.	YOLO or suitable computer-vision model + OpenCV
Severity/safety assessment	Generate a visual safety/severity indicator from detected hazards and evidence.	Computer vision + calibrated scoring rules
Priority prediction	Predict urgency using contextual and historical features.	XGBoost or Random Forest
Department routing	Map issue descriptions/categories to	NLP/classification + configurable

	the appropriate maintenance division.	routing rules

6. Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Student, staff, and administrator interfaces
UI	Tailwind CSS	Responsive interface development
Backend	Python + FastAPI	REST API, authentication, business logic, AI integration
Database	PostgreSQL	Users, reports, tasks, departments, history, notifications
AI/ML	Python, PyTorch, scikit-learn	Model development and inference
Computer Vision	OpenCV + YOLO/appropriate vision model	Image processing and hazard detection
NLP	Transformers	Text classification and language processing
Similarity	Embeddings + cosine similarity	Duplicate-ticket detection
Priority ML	XGBoost / Random Forest	Maintenance urgency prediction
Testing	Postman + unit/integration tests	API and system testing
Development	VS Code + Jupyter	Application and model development
Version Control	Git + GitHub	Source-code management

7. Database Design

```
USERS: user_id, name, student_id, email, role, password_hash

REPORTS: report_id, user_id, location_id, description, image_url,
category, status, created_at

AI_ANALYSIS: analysis_id, report_id, category, confidence, hazard,
severity_score, duplicate_score, priority, department

DEPARTMENTS: department_id, name, description

MAINTENANCE_TASKS: task_id, report_id, department_id, staff_id,
priority, status, assigned_at, started_at, completed_at

STATUS_HISTORY: history_id, report_id, status, changed_by, timestamp

NOTIFICATIONS: notification_id, user_id, report_id, message,
is_read, created_at

FEEDBACK: feedback_id, report_id, user_id, rating, comment, created_at
```

8. Student UI Wireframe – Login

```
+--------------------------------------+
|         CAMPUS MAINTENANCE           |
|                                      |
|              Student Login           |
|                                      |
| Email / Student ID                   |
| [____________________________]       |
|                                      |
| Password                             |
| [____________________________]       |
|                                      |
|              [   LOGIN   ]           |
+--------------------------------------+
```

9. Student UI Wireframe – Dashboard

```
+----------------------------------------------+
| CAMPUS MAINTENANCE              [Bell] [User] |
+----------------------------------------------+
| Welcome, Student                             |
|                                              |
|           [ + REPORT A PROBLEM ]             |
|                                              |
| MY REPORTS                                   |
| +------------------------------------------+ |
| | Broken AC | Building A - Room 301        | |
| | Status: IN PROGRESS                      | |
| +------------------------------------------+ |
```

```
| +-----------------------------------------+ |
| | Broken Light | Building B - Room 204    | |
| | Status: RESOLVED                        | |
| +-----------------------------------------+ |
+---------------------------------------------+
```

10. Student UI Wireframe – Report Issue

```
+---------------------------------------------+
|            REPORT CAMPUS ISSUE              |
+---------------------------------------------+
| Upload Photo                                |
|          +----------------------+           |
|          |      [ UPLOAD ]      |           |
|          +----------------------+           |
| Building: [ Engineering Building v ]        |
| Floor:    [ 3 v ]                           |
| Room:     [ 301 ]                           |
| Describe the problem                        |
| +-----------------------------------------+ |
| | There is water leaking from the AC...   | |
| +-----------------------------------------+ |
|             [ SUBMIT REPORT ]               |
+---------------------------------------------+
```

11. AI Analysis Screen Wireframe

```
+---------------------------------------------+
|                 AI ANALYSIS                 |
+---------------------------------------------+
| Issue Category: HVAC / Air Conditioning     |
| Confidence: 94%                             |
| Safety Assessment: 7.8 / 10                 |
| Hazard Detected: Water Leakage              |
| Duplicate Check: No matching active incident |
| Predicted Priority: HIGH                    |
| Assigned Department: HVAC Maintenance       |
+---------------------------------------------+
```

12. Maintenance Staff Dashboard Wireframe

```
+---------------------------------------------+
| MAINTENANCE PORTAL                 [Staff]  |
+---------------------------------------------+
| ASSIGNED TASKS                              |
| +-----------------------------------------+ |
| | CRITICAL                                | |
| | Exposed Electrical Wiring               | |
| | Engineering Building - Room 301         | |
| | Safety Risk: HIGH                       | |
| | [ ACCEPT ]       [ VIEW REPORT ]        | |
| +-----------------------------------------+ |
| | MEDIUM                                  | |
| | Broken Classroom Light                  | |
| | Building B - Room 204                   | |
| | [ ACCEPT ]       [ VIEW REPORT ]        | |
| +-----------------------------------------+ |
+---------------------------------------------+
```

13. Administrator Dashboard Wireframe

```
+---------------------------------------------+
|               ADMIN DASHBOARD               |
+---------------------------------------------+
| ACTIVE ISSUES     CRITICAL      RESOLVED    |
|     27               3            142       |
+---------------------------------------------+
| ISSUES BY CATEGORY                          |
| Electrical       ################           |
| HVAC             ############               |
| Plumbing         ########                   |
| IT               ######                     |
| Other            ####                       |
+---------------------------------------------+
| DEPARTMENT WORKLOAD                         |
| Electrical       8 active                   |
| HVAC             6 active                   |
| Plumbing         5 active                   |
| IT               4 active                   |
+---------------------------------------------+
| Escalated Issues: 3                         |
```

```
| Recurring Problems: 7                          |
+------------------------------------------------+
```

14. Automatic Escalation Workflow

```
Critical / High-Priority Issue
             |
             v
     Assigned to Department
             |
             v
     Response Timer Starts
             |
        No response?
             |
             v
      Automatic Reminder
             |
        Still delayed?
             |
             v
   Administrator Notification
             |
             v
      Task Escalated
```

Response targets should be configurable by priority. Final response times should be confirmed with the university before implementation.

15. Recommended Development Strategy

Phase	Main Work	Outcome
1	Requirements, architecture, database, authentication, reporting workflow	Working non-AI reporting system
2	AI baseline + database development in parallel	Initial AI analysis pipeline
3	Multimodal classification + duplicate detection	Intelligent issue understanding
4	Computer vision hazard/severity + priority ML	Risk-aware maintenance prioritization
5	Department routing + staff dashboard + escalation	Automated maintenance dispatch
6	Analytics, recurring-problem detection, testing and evaluation	Final integrated capstone system

16. 15-Week Development Plan

Weeks	Activities
1–2	Requirements analysis, system architecture, dataset and evaluation planning
2–4	Database development and AI baseline modeling concurrently
3–6	Backend/API, authentication, reporting workflow and React interfaces
5–8	Multimodal issue classification and duplicate-ticket detection
7–10	Computer-vision hazard detection and severity/safety assessment
9–11	ML priority prediction and automated department routing
10–12	Maintenance dispatch, status tracking, notifications and automatic escalation
12–13	Administrator analytics and recurring-problem detection
13–14	Full integration, debugging, usability/security testing and model evaluation
15	Final optimization, documentation, demonstration and presentation

17. Scope Control

The project should avoid unnecessary hardware dependencies such as QR labels, IoT sensor networks, CCTV monitoring, facial recognition, drones, or a custom large language model. The core deliverable is a software-based AI/ML maintenance workflow. Advanced features should be implemented only after the core reporting, AI analysis, dispatch, escalation, and resolution workflow is stable.

18. Expected Final Demonstration

A complete demonstration can follow one incident: a student uploads a photograph and description; the AI classifies it; the system checks for duplicates; computer vision assesses visible safety risk; the ML model predicts priority; the issue is routed to the appropriate maintenance division; staff accept and update the task; the escalation mechanism is demonstrated using a configurable timeout; the student receives status updates; and the administrator views the incident and analytics.